DANIEL ANCHUELA CANTARERO

AI ENGINEER | GENERATIVE AI & COMPUTER VISION

28012, Madrid, Spain

(+34) 699478216 | dani.anchuela09@gmail.com | LinkedIn | Portfolio

ABOUT

AI Engineer with a background in Biomedical Engineering and a specialization in Data Science, experienced in developing end-to-end AI solutions across generative AI, computer vision, and applied machine learning. My work spans enterprise GenAI systems for knowledge retrieval and document intelligence, large-scale image analysis pipelines, and data-driven workflows for complex signal and multimodal data. I combine strong technical execution with a practical, product-oriented mindset, building scalable solutions that turn complex data into actionable insights.

EDUCATION

ROVIRA I VIRGILI UNIVERSITY — **TARRAGONA, SPAIN**

Master's in Biomedical Data Science — 2023 – 2025

SAN PABLO CEU UNIVERSITY — **MADRID, SPAIN**

Bachelor's Degree in Biomedical Engineering — 2019 – 2023

PROFESSIONAL EXPERIENCE

ASTRAZENECA'S RARE DISEASE UNIT — **BARCELONA, SPAIN**

Artificial Intelligence Engineer — 2024 – PRESENT

- ***Enterprise GenAI Projects:*** *Led the design and delivery of enterprise GenAI solutions for knowledge retrieval and document intelligence, translating business requirements into production-ready AI systems.*
- ***Document Ingestion & ETL:*** *Built and deployed a scalable document ingestion and ETL pipeline for enterprise content, covering multimodal extraction, orchestration, and downstream data connectivity.*
- ***Platform Integrations & Observability:*** *Integrated key platform capabilities and enterprise connectors, including Microsoft Graph API, SharePoint, and LangSmith, to support real-world ingestion, observability, and system validation.*
- ***MLOps & AWS Deployment:*** *Supported MLOps-related deployment and validation of AI solutions in AWS UAT environments, contributing to pipeline workflows, monitoring, and production readiness.*
- ***Architecture & Collaboration:*** *Contributed to architecture decisions, technical documentation, and stakeholder communication across technical and business teams.*

SPANISH NATIONAL CANCER RESEARCH CENTER (CNIO) — **MADRID, SPAIN**

Computer Vision Engineer — 2024 – 2025

- ***Microscopy Image Analysis Pipeline:*** *Developed a modular Python pipeline for multiplex microscopy image analysis, including segmentation, feature extraction, cell phenotyping, and spatial analysis at single-cell resolution.*
- ***Tumor Microenvironment Quantification:*** *Built automated workflows to quantify tumor heterogeneity, immune infiltration, and cell–cell proximity from multichannel fluorescence images.*
- ***Deep Learning for Histopathology:*** *Developed and validated a CNN-based attention-enhanced U-Net for tumor invasion front detection in H&E images.*
- ***HPC/GPU Data Processing:*** *Processed large-scale microscopy datasets using HPC/GPU resources to optimize training and inference performance.*
- ***Translational Research Collaboration:*** *Collaborated with multidisciplinary oncology teams to translate imaging data into actionable biological insights.*

WERIUM SOLUTIONS **MADRID, SPAIN**

Data Scientist and Researcher 2022 – 2024

- ***Biomedical Signal Analysis**: Processed EMG and ROM biosignals using filtering, normalization, and artifact removal techniques.*
- ***Biomechanical Metrics Extraction**: Computed biomechanical metrics including coactivation index, joint range of motion, gait cadence, step width, and gait phase segmentation.*
- ***Statistical Analysis**: Performed comparative analyses, identified significant changes, and generated tables and visual outputs to support result interpretation.*

SCIENTIFIC PUBLICATIONS, CONFERENCES & PERSONAL PROJECTS

- ***Agentic computer vision chatbot for astronomical image analysis (2026)***
- ***Spatial Biology Computer Vision Tool for Multiplex Microscopy (2025)*** - *View project*
- ***Easton Royal – Virtual Reality Interactive Game (2025)*** - *View project*
- ***Leveraging Nextflow for Scalable Processing in Fluorescence Microscopy (2024)*** - *Presented at Spanish & Portuguese Advanced Optical Microscopy Meeting 2024* - *View project*
- ***Gait alterations in older adults walking with different body weight support and velocity settings on the SWalker robotic platform (2024)*** - *Scientific publication in progress*
- ***Detection and classification of heartbeats with neural networks (2023)*** - *View project*

MAIN SKILLS

- **Programming:** *Python, SQL, R, Bash, Java*
- **Machine Learning:** *PyTorch, TensorFlow, Scikit-learn, supervised learning, deep learning, model evaluation*
- **Generative AI:** *RAG, embeddings, semantic search, multimodal retrieval, intelligent document processing, generative BI, LangChain, LangSmith*
- **Computer Vision:** *OpenCV, CNNs, U-Net, image classification, image segmentation, feature extraction, quantitative image analysis*
- **Data Engineering & Systems:** *ETL pipelines, workflow orchestration, API integration, scalable data processing*
- **MLOps & Cloud (AWS):** *SageMaker, S3, ECS, CodeBuild, CodePipeline, CloudWatch, deployment validation, monitoring*
- **Analytics & Signal Processing:** *Filtering, denoising, Fourier analysis, feature extraction, statistical analysis, hypothesis testing, data visualization, SciPy*
- ***Other tools:** Git, Docker, Linux, FastAPI, PostgreSQL, Redis, Celery, Alembic, HPC clusters*

LANGUAGES

Spanish: Native; **Catalan:** Native; **English:** C1;

ADDITIONAL EDUCATION AND INFORMATION

- Master's degree in Astronomy and Astrophysics
- Master's degree in Virtual Reality development
- Former semi-professional football player